# Hybrid Quantum AI Models for Accelerating Multiphysics Simulation

Hybrid-QPINN / QPINO: Fourier features, data re-uploading and shallow parameterised quantum circuits

Aswin

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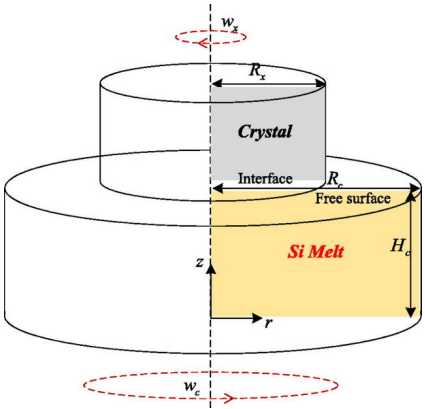
# Outline

# Accurate multiphysics simulation is too slow for design loops

- High-fidelity coupled CFD (flow + heat + rotation + phase change) takes hours to days per run.
- The real cost is the **parameter sweep**: every operating point is a new simulation.
- Process control and digital twins need full fields in **milliseconds**.
- Classical surrogates (PINN, PINO) help, but stall on sharp, strongly coupled fields.

## Goal

A surrogate that keeps the governing physics, generalises across operating conditions, and evaluates in real time.

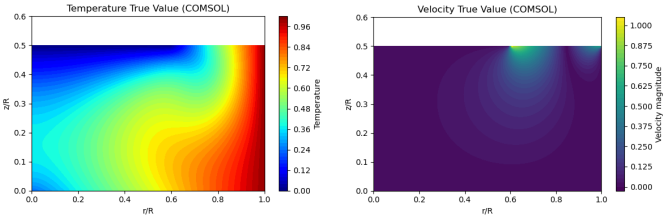



Czochralski (CZ) growth: rotating crystal ($\omega_s$) pulled from a Si melt in a counter-rotating crucible ($\omega_c$).

# Two industrial use cases

## Silicon crystal growth (Czochralski)

Melt convection, swirl and the thermal field (1685–1750 K) set defect density and oxygen uptake. **Aim:** fast field prediction to tune the process.




COMSOL reference: temperature (left) and velocity magnitude (right) in the melt. Sharp gradients sit at the free surface and crystal edge.

## $LH_2$ cryogenic boil-off

- Heat leak into liquid-hydrogen tanks (≈20 K) drives natural convection, stratification and evaporation.
- Turbulent convection with phase change and a **moving liquid–vapour interface**.
- **Aim:** predict boil-off rate and pressure rise for tank design.

## Common thread

Strongly coupled, non-linear transport of velocity, temperature, species and interfaces.

# Our proposal in one slide

1. **Problem:** classical PINNs suffer from spectral bias, loss imbalance and badly scaled coupled fields.
2. **Framework:** Hybrid-QPINN / QPINO, a classical network with a shallow parameterised quantum circuit (PQC) as a trainable feature map, fed by Fourier features and data re-uploading.
3. **Target physics:** strongly coupled non-linear transport: velocity, temperature, and next concentration and evolving interfaces.
4. **Benchmarks:** silicon CZ crystal growth (running) and $LH_2$ boil-off (defined).
5. **Destination:** real-time digital twins and fast parametric exploration.

## How to read the results

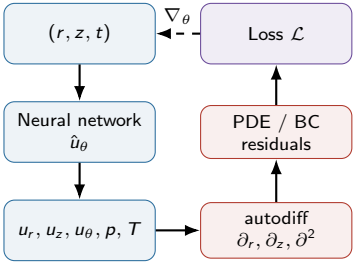
Each claim carries a status: **[Shown]** measured, **[In progress]** experiment running, **[Planned]** designed but not yet run.

# Physics-informed neural networks in one slide

- A network $\hat{u}_\theta(x)$ is trained so that the PDEs and boundary/initial conditions hold.
- Derivatives come from automatic differentiation: no mesh, no stencil.

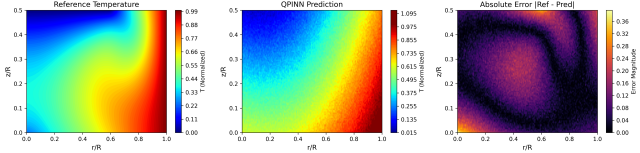
$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}}$$

- Strength: little labelled data, natural fit for coupled equations.
- Limitation: one PINN = one operating point. Neural operators (PINO) learn whole solution families.



# Bottleneck 1: spectral bias

- MLPs learn low frequencies first; sharp content converges slowly (Rahaman et al., 2019).
- Transport fields are full of sharp structure: thermal boundary layers, wall shear, swirl near rotating walls, phase interfaces.
- Classical remedy: Fourier feature inputs (Tancik et al., 2020).
- **Our addition:** a *trainable* quantum feature map whose outputs are trigonometric polynomials of its inputs.



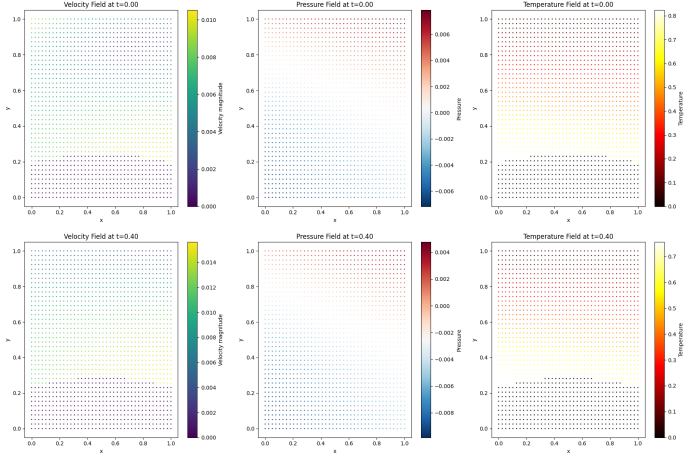

Reference temperature, a smooth network prediction, and the absolute error. The error concentrates on the curved thermal front, exactly where the field carries high-frequency content.

# Classical PINN baseline on simplified crystal growth




Training loss: total, PDE and BC/IC components (50 epochs).

- Total loss is driven by BC/IC; the PDE term oscillates rather than falling steadily.
- Loss ends rising again after epoch 45.




Predicted velocity, pressure and temperature at $t = 0$ (top) and $t = 0.4$ (bottom). Fields stay diffuse; the near-interface layer is barely resolved.

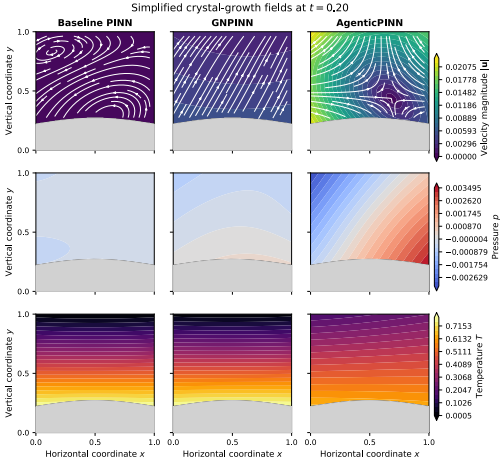
# Bottleneck 2: loss-term imbalance

| Trainer (1D heat) | PDE loss | BC/IC loss | Rel. $L_2$ |
|---|---|---|---|
| Baseline | $3.96\times10^{-3}$ | $1.15\times10^{-3}$ | 0.084 |
| GNPINN | $9.74\times10^{-3}$ | $9.05\times10^{-4}$ | 0.054 |
| AgenticPINN | $8.22\times10^{-7}$ | $2.00\times10^{-1}$ | 1.368 |

- The term with the largest raw gradient dominates (Wang et al., 2021).
- AgenticPINN: PDE residual ≈4800× smaller, BC/IC ≈173× larger, and the field is wrong.

## Lesson carried into this work

Report loss components and per-field errors, never only a weighted total.




Simplified CZ fields at $t = 0.2$: baseline PINN, GNPINN, AgenticPINN (velocity, pressure, temperature). Chapagain et al., 2026.

# Bottleneck 3: scale mismatch between coupled fields

- Five coupled outputs: $u_r, u_z, u_\theta, p, T$, with very different magnitudes.
- In the CZ model the pressure coefficient is

$$p_{\text{coeff}} = \frac{p_{\text{scale}}}{v_{\text{scale}}^2} = 134{,}128.$$

- Viscous terms scale as $1/Re \approx 0.067$.
- Pressure-to-viscous ratio $\approx 2 \times 10^6 : 1$.

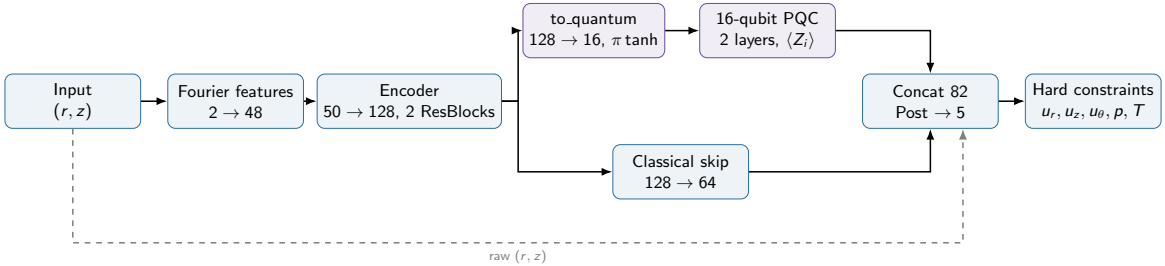
**Consequence**

The pressure residual can overwhelm the data signal during physics-informed training. This is the main stability issue we measured (Result 3).

**Design response**

Hard constraints, staged physics warm-up, EMA residual balancing, and next per-field residual scaling.

# Architecture overview




## Classical backbone

Fourier encoding plus residual MLP: carries most of the computation.

## Quantum feature map

16 qubits, 2 layers, 96 trainable angles; 16 expectation values out.

## Skip path

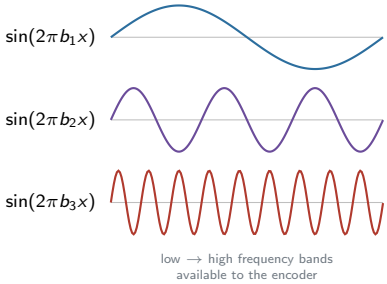
The head never depends solely on the quantum bottleneck.

# Ingredient 1: Fourier feature mapping

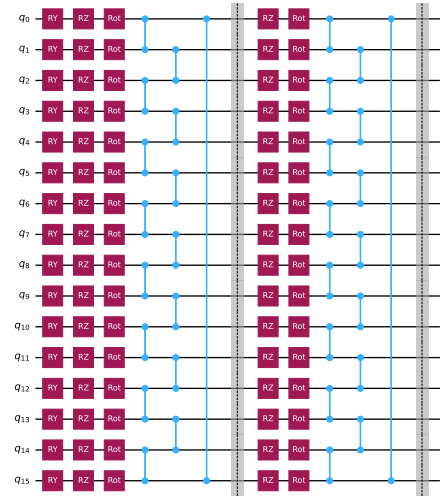
- Frozen random matrix $B \in \mathbb{R}^{2\times 24}$, $B_{ij} \sim \mathcal{N}(0, 6^2)$.

$$\gamma(x) = \big[\sin(2\pi\, xB),\ \cos(2\pi\, xB)\big] \in \mathbb{R}^{48}$$

- Frequencies span roughly $\pm 15$ cycles per unit length in normalised coordinates.
- Concatenated with raw $(r, z)$: 50 inputs to the encoder.
- High-frequency access from the first layer, at **zero trainable cost**.



# Ingredient 2: shallow parameterised quantum circuit




Logical circuit, 16 qubits $\times$ 2 layers.

1. **Encode:** $R_Y(\theta_i)$, $\theta_i = \pi \tanh(h_i) \in [-\pi, \pi]$.
2. **Re-upload:** $R_Z(0.5\,\theta_{(i+\ell) \bmod 16})$.
3. **Train:** $\mathrm{Rot}(\varphi_i, \vartheta_i, \omega_i)$, 48 angles per layer.
4. **Entangle:** CZ ring, closing $q_{15}$–$q_0$.

## Readout

$\langle Z_i \rangle$, $i = 0, \ldots, 15 \rightarrow$ LayerNorm(16).
**96 trainable quantum parameters**, about 50 gates per shot on hardware.

# Ingredient 3: data re-uploading

- A PQC output is a truncated Fourier series of its encoded input (Schuld et al., 2021):

$$f_\phi(\theta) = \sum_{\omega \in \Omega_L} c_\omega(\phi)\, e^{i\omega\theta}$$

- Encoding once gives a narrow spectrum $\Omega_1$; re-uploading in every layer widens it to $\Omega_L$ (Pérez-Salinas et al., 2020).
- The cyclic shift $\theta_{(i+\ell) \bmod 16}$ lets each layer mix a different combination of latent features.
- With Fourier inputs, this targets exactly the content that spectral bias suppresses.

**Design rule**

The PQC is a compact, expressive *feature extractor*. The classical network carries the bulk of the computation, which keeps the circuit shallow enough for today's hardware.

# Parameter budget: a deliberately small quantum layer

| Component | Parameters |
|---|---|
| Coordinate projector | 23,040 |
| ResidualMLP blocks ($2 \times 2$) | 66,048 |
| to_quantum | 9,296 |
| Classical skip | 8,256 |
| **Quantum circuit** | **96** |
| Quantum LayerNorm | 32 |
| Post network | 19,205 |
| Fourier $B$ (frozen) | 48 |
| **Total** | **126,021** |

## Why small?

- Shallow circuits survive NISQ noise.
- Few angles $\Rightarrow$ cheap hardware fine-tuning.
- 0.076% of all parameters.

## Consequence for evaluation

Any gain must be attributed to the PQC by a controlled ablation, not assumed.

# Governing equations: coupled flow, swirl and heat

Steady, axisymmetric, incompressible Navier–Stokes with Boussinesq buoyancy (nondimensional):

$$\text{mass:} \quad \partial_r(r u_r) + r\,\partial_z u_z = 0$$

$$r\text{-mom.:} \quad u_r u_{r,r} + u_z u_{r,z} - \frac{u_\theta^2}{r} + p_{\text{coeff}}\, p_{,r} - \frac{1}{Re}\Big[u_{r,rr} + \frac{u_{r,r}}{r} - \frac{u_r}{r^2} + u_{r,zz}\Big] = 0$$

$$z\text{-mom.:} \quad u_r u_{z,r} + u_z u_{z,z} + p_{\text{coeff}}\, p_{,z} - \frac{1}{Re}\Big[u_{z,rr} + \frac{u_{z,r}}{r} + u_{z,zz}\Big] - \frac{Gr}{Re^2}\,\theta = 0$$

$$\text{swirl:} \quad u_r u_{\theta,r} + u_z u_{\theta,z} + \frac{u_r u_\theta}{r} - \frac{1}{Re}\Big[u_{\theta,rr} + \frac{u_{\theta,r}}{r} - \frac{u_\theta}{r^2} + u_{\theta,zz}\Big] = 0$$

$$\text{energy:} \quad u_r \theta_{,r} + u_z \theta_{,z} - \frac{1}{Pr\,Re}\Big[\theta_{,rr} + \frac{\theta_{,r}}{r} + \theta_{,zz}\Big] = 0$$

$$\theta = (T - T_{\min})/(T_{\max} - T_{\min}), \quad Re = 15, \quad Pr = 28.463, \quad Gr = 8000, \quad p_{\text{coeff}} = 134{,}128$$

Non-linear two-way coupling: swirl feeds $r$-momentum via $u_\theta^2/r$, temperature feeds $z$-momentum via buoyancy.

# Physics loss and hard constraints

- Derivatives via `torch.autograd.grad`, `create_graph=True` (second order).
- Five residuals (continuity, $r$, $z$, swirl, energy), balanced by EMA weighting, $\beta = 0.95$.
- Staged schedule: data only until epoch 50, then physics weight ramped $0 \to 0.05$.
- Gradient clipping at L2 norm 1.0.
- **Hard constraints:** $u_r = r \cdot \mathrm{raw}_0$, $u_\theta = r \cdot \mathrm{raw}_2$, so both vanish exactly on the axis.

**Current limitation**

The PQC output is detached inside the physics loss, so PDE gradients do not reach the quantum angles. The PQC is shaped by the data loss only.

**Fix in progress [In progress]**

Parameter-shift or adjoint gradients for the residual path.

# From QPINN to QPINO: learning across operating conditions

## QPINN (today)

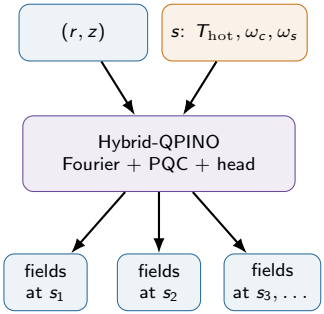
One model per operating condition:
$(r, z) \mapsto (u_r, u_z, u_\theta, p, T)$.

## QPINO (next)

Process parameters become inputs:

$$(r, z, s) \mapsto (u_r, u_z, u_\theta, p, T), \quad s \in \{T_{\mathrm{hot}}, \omega_c, \omega_s, v_{\mathrm{pull}}\}$$

One trained surrogate serves the whole sweep: the basis for parametric exploration and digital twins.



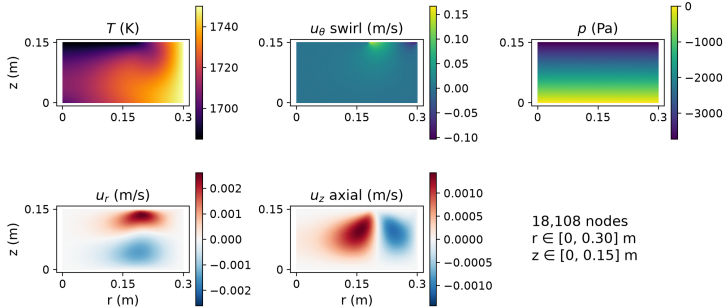

one training run, many conditions, ms-scale queries

# Benchmark 1: silicon CZ melt [In progress]

| Property | Value |
|---|---|
| Reference | Axisymmetric CFD |
| Domain | $r \in [0, 0.30]$ m |
| | $z \in [0, 0.15]$ m |
| Nodes | 18,108 |
| Outputs | $u_r, u_z, u_\theta, p, T$ |
| $T$ range | 1685–1750 K |
| Split | 17,203 / 905 |

## Scope

Steady, laminar, single-phase reference; geometry and BCs follow Huang et al. (2025).




Training data: $T$, swirl $u_\theta$, $p$, radial $u_r$ and axial $u_z$. Note the counter-rotating meridional cells and the thin swirl layer under the crystal.

## Next physics

Oxygen concentration transport (species equation) and the evolving melt–crystal interface.

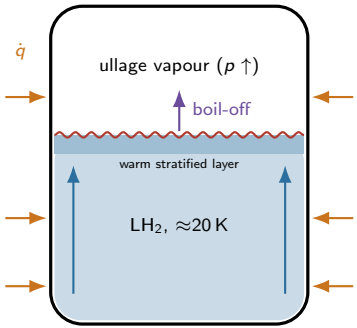
# Benchmark 2: $LH_2$ cryogenic boil-off [Planned]

## Physics to capture

- Wall heat leak → natural-convection boundary layer and thermal stratification.
- Turbulent convection (RANS closure in the reference CFD).
- Liquid–vapour interface with evaporation (VOF + phase-change mass transfer).
- Outputs: velocity, temperature, phase fraction, boil-off rate, ullage pressure.

## Status

Benchmark and outputs defined; reference CFD data not yet generated.

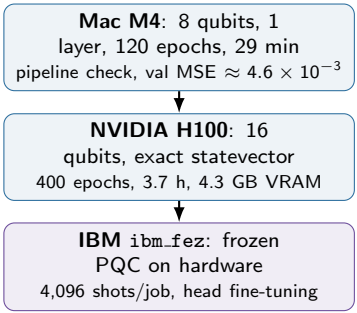



moving interface + natural convection

# Evaluation protocol and compute pipeline

- **Ablations:** classical-only at matched budget; random-feature layer in place of the PQC.
- **Baselines:** classical PINN and PINO, matched optimiser, sampling, seeds.
- **Generalisation:** hold out a whole operating condition.
- **Reporting:** per-field error, unweighted loss components.
- **Cost:** wall time to target error, inference latency.

| Sweep | Range | Held out |
|---|---|---|
| $T_{\mathrm{hot}}$ | 1730–1785 K | 1780 K |
| $\omega_c$ (crucible) | $-10$ to $-1$ rpm | $-4$ rpm |
| $\omega_s$ (crystal) | 4–20 rpm | 7 rpm |




**Note**

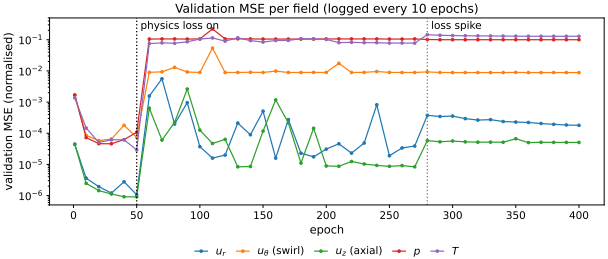
16 qubits $= 2^{16}$ amplitudes: exactly simulable on one GPU. Hardware runs test noise robustness and deployment, not quantum advantage.

# Result 1: fast convergence in the data-driven phase **[Shown]**

| Epoch | Train loss | Val MSE | Val $T$ |
|---|---|---|---|
| 1 | $1.16\times10^{-2}$ | $4.83\times10^{-3}$ | $1.37\times10^{-3}$ |
| 10 | $5.92\times10^{-5}$ | $3.11\times10^{-4}$ | $1.47\times10^{-4}$ |
| **20** | $5.26\times10^{-5}$ | $1.58\times10^{-4}$ | $5.07\times10^{-5}$ |
| 30 | $2.96\times10^{-5}$ | $1.75\times10^{-4}$ | $6.15\times10^{-5}$ |
| 50 | $2.07\times10^{-5}$ | $2.13\times10^{-4}$ | $2.91\times10^{-5}$ |

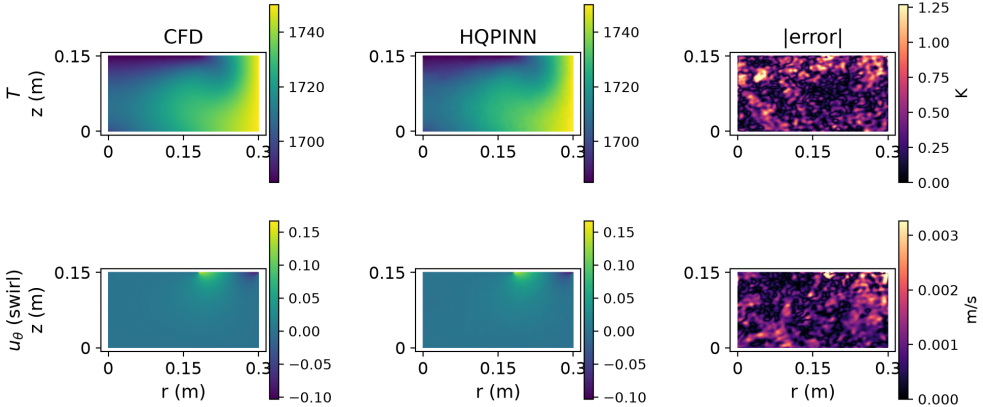
## Reading

Validation MSE drops **30× in 20 epochs**. Best checkpoint $1.58\times10^{-4}$ (random spatial split, in distribution).




Per-field validation MSE (H100, 400 epochs). Fields reach $10^{-6}$–$10^{-4}$ before the physics loss switches on at epoch 50.

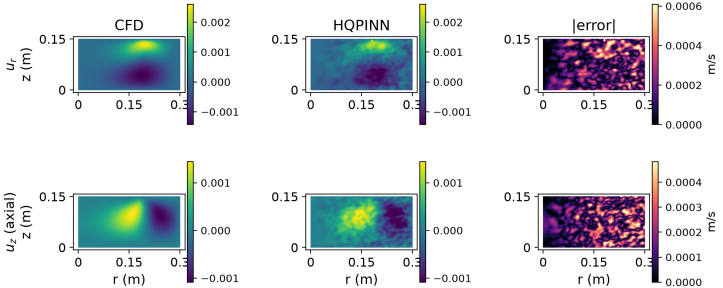
# Result 2: thermal and swirl fields captured **[Shown]**




CFD reference vs Hybrid-QPINN (best checkpoint) and absolute error: temperature $T$ (top, error $\lesssim$1.25 K on a 65 K range) and swirl $u_\theta$ (bottom, error $\lesssim 3 \times 10^{-3}$ m/s on $\pm$0.15 m/s).

The thin swirl layer under the rotating crystal and the hot-wall thermal gradient are both reproduced.

# Result 2 (cont.): meridional velocities, the hardest fields **[Shown]**




$u_r$ (top) and $u_z$ (bottom): CFD, Hybrid-QPINN, |error|.

| Field | Val MSE (norm.) |
|---|---|
| $u_r$ | $1.93 \times 10^{-6}$ |
| $u_z$ | $5.78 \times 10^{-5}$ |
| $u_\theta$ | $1.44 \times 10^{-6}$ |
| $p$ | $4.62 \times 10^{-5}$ |
| $T$ | $5.07 \times 10^{-5}$ |

- Counter-rotating cells located correctly.
- Errors are small-scale and noisy: residual high-frequency content still to be learned.

# Result 3: naive physics weighting destabilises training [Shown]

| Field | Best (ep. 20) | Final (ep. 400) | Factor |
|---|---|---|---|
| $u_r$ | $1.93 \times 10^{-6}$ | $1.79 \times 10^{-4}$ | 93× |
| $u_z$ | $5.78 \times 10^{-5}$ | $8.83 \times 10^{-3}$ | 153× |
| $u_\theta$ | $1.44 \times 10^{-6}$ | $5.06 \times 10^{-5}$ | 35× |
| $p$ | $4.62 \times 10^{-5}$ | 0.100 | **2,165**× |
| $T$ | $5.07 \times 10^{-5}$ | 0.129 | **2,544**× |

```
[0280] loss=1.04e+09 data=5.06e-02 phys=2.16e+10
[0290] loss=1.18e+14 data=4.97e-02 phys=2.15e+15

[0300] loss=5.57e-02 data=4.85e-02 phys=9.62e-02
```

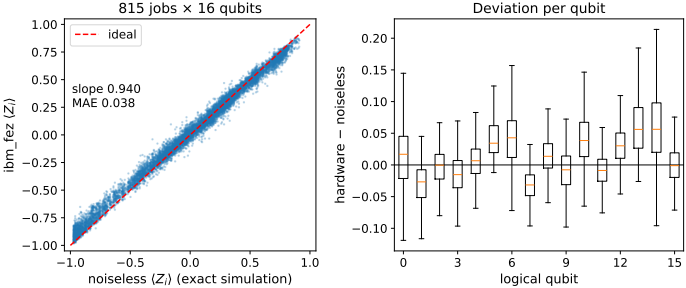
## Diagnosis

Data loss stays near 0.05; the blow-up is entirely in the physics residual (jump at epoch 50, spike at 280 in Result 1's plot). Root cause: the $p_{\text{coeff}}$ scale mismatch.

## Fixes under test [In progress]

Physics weight 0.001–0.005; per-field pressure residual scaling; re-check pressure nondimensionalisation.

# Result 4: the circuit runs on real quantum hardware [Shown]



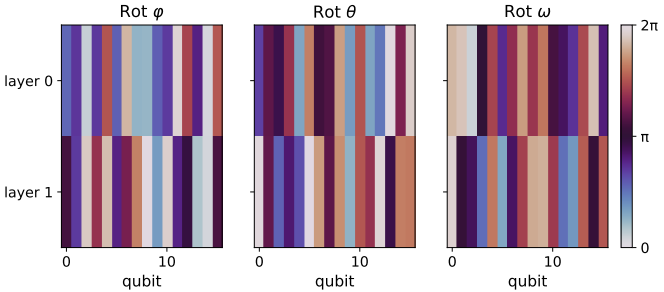

ibm_fez vs exact simulation, 815 jobs × 16 qubits: slope 0.940, MAE 0.038; per-qubit deviation stays within about ±0.1.

| Fine-tuning session | Value |
|---|---|
| Backend | ibm_fez (156 q) |
| Jobs completed / failed | **500 / 0** |
| Shots per job | 4,096 |
| Total shots | 2,048,000 |
| Session | 125 min |
| Trainable scope | head (19,205) |

## Reading

Hardware features are a mildly compressed (slope $< 1$), low-bias copy of the simulated ones.
Strategy: freeze the PQC, re-adapt the classical head on hardware $\langle Z_i \rangle$. **[In progress]**

# Inside the trained circuit




Trained Rot angles $(\varphi, \vartheta, \omega)$ per qubit, layers 0 and 1.

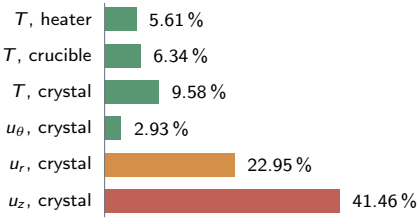
- Angles span the full $[0, 2\pi]$ range (mean 3.49, std 1.79 rad).
- No qubit stuck at a degenerate value; the two layers learn different patterns.
- In a longer run, LayerNorm weights qubits unequally ($\gamma \approx 0.3$–$1.2$): the head relies on some quantum features more than others.

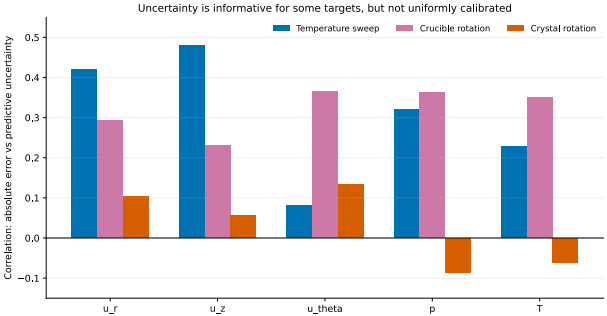
The circuit is active and differentiated. Whether it adds accuracy is what the ablation will decide.

# Parametric exploration: what a classical surrogate gives today

## GPR baseline, held-out case

Learns $(r, z, s) \mapsto$ fields; NRMSE = RMSE / std of the field.




Correlation between GPR predictive uncertainty and actual error, per field and sweep.

$T$ generalises (5.6–9.6%); meridional velocities under crystal rotation do not (23–41%). This is the bar QPINO must beat.

# Where the evidence stands today

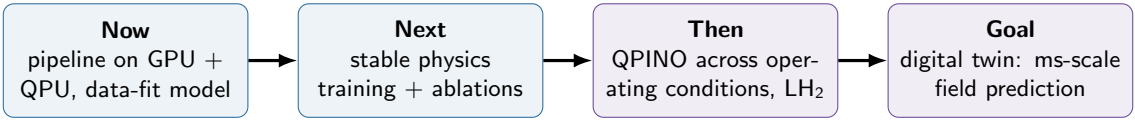
| Claim | Status | Evidence |
|---|---|---|
| Hybrid pipeline runs on GPU and real QPU | [Shown] | 500/500 IBM jobs; HW vs sim MAE 0.038 |
| Five coupled CZ fields fitted in distribution | [Shown] | Val MSE $1.58 \times 10^{-4}$ |
| Fast early convergence | [Shown] | $30\times$ drop in 20 epochs |
| Physics loss needs field-aware scaling | [Shown] | $2{,}000\times$ degradation for $p$, $T$ |
| Faster convergence than classical PINN/PINO | [In progress] | Matched baseline running |
| PQC adds accuracy over classical features | [In progress] | Ablation + random-feature control |
| Sharper generalisation across conditions | [Planned] | Case-wise holdout vs GPR / PINO |
| Reduced computational cost | [Planned] | GPU-hours to target error |
| Concentration, evolving interfaces, $LH_2$ | [Planned] | Species eq., interface tracking |

Early indications are positive; the hybrid-vs-classical advantage is decided by the matched ablations.

# Next experiments

| Priority | Experiment | Compute |
|---|---|---|
| Critical | Classical-only baseline, matched parameters | $1\times$H100, $\sim$4 h |
| Critical | Random-feature control in place of the PQC | $1\times$H100, $\sim$4 h |
| Critical | Retrain with physics weight 0.001–0.005 | $1\times$H100, $\sim$4 h |
| High | Physics gradients through the PQC (parameter shift) | architecture change |
| High | Case-wise holdout on three CZ sweeps (QPINO) | 3 runs |
| High | Complete head fine-tuning on IBM hardware | $\sim$500 jobs |
| Medium | $LH_2$ boil-off reference CFD + surrogate | CFD + H100 |

# Pathway to real-time digital twins




## Data engine

Agentic CFD (AutoFOAM) generates verified training cases from one prompt.

## Surrogate

Hybrid-QPINO trained once, queried across the full process window for parametric exploration.

## Validator

Physics residuals, uncertainty and benchmark checks before any prediction is trusted.

# Key takeaways

1. **Framework:** Fourier features + data re-uploading + a shallow 16-qubit PQC (96 angles) as a trainable feature map inside a physics-informed network.
2. **Demonstrated:** all five coupled CZ fields fitted (val MSE $1.58 \times 10^{-4}$, $30\times$ drop in 20 epochs); the circuit runs on IBM hardware (500/500 jobs, MAE 0.038 vs simulation).
3. **Learned:** physics weighting must respect field scales; a pressure coefficient of $\sim 10^5$ can wreck training.
4. **Being tested:** faster convergence, sharper generalisation and lower cost than classical PINN/PINO, via matched ablations.
5. **Destination:** QPINO surrogates for crystal growth and $LH_2$ storage as the core of real-time digital twins.

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Questions?

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